

PROFILE

- Full-stack software engineer experienced in production-grade AI systems: RAG pipelines, fraud-detection engines, OCR automation.
- Proficient in Java (DSA), Python, C++, C; strong OOP and System Design fundamentals.
- Skilled across full stack (Next.js, FastAPI, Node.js, PostgreSQL) and AI/ML layers (LLMs, embeddings, YOLO, EasyOCR).
- Focused on scalability, cost-efficient architecture, and real-world usability.

PROJECTS

Fin-Sight — Financial RAG Platform

github.com/Debarghyasg/Fin-Sight[Next.js 15](#) · [FastAPI](#) · [PostgreSQL](#) · [MiniLM](#) · [Groq Llama-3.3-70B](#) · [Celery](#) · [Presidio](#) · [FinBERT](#) · [Docker](#) · [Git](#)

- Built multi-tenant RAG platform for SEC 10-K filings: cited Q&A, anomaly alerts, live EDGAR watch. CI/CD via GitHub Actions, \$0/month dev stack.
- Designed zero-extension vector store: PostgreSQL JSON embeddings + NumPy cosine search, replacing Qdrant/Pinecone.
- Implemented adaptive chunking (800-char windows + table preservation), cutting citation errors from ~30% to near-zero.
- Achieved <660ms P95 query latency on CPU; ~6.6× cheaper than GPT-4o + Pinecone at 1,000 users (\$386 vs \$2,563/month).

Nyatik Nayan — Retail Fraud Detection Platform

github.com/Debarghyasg/Nyatik-Nayan[React 19](#) · [Node.js/Express](#) · [FastAPI](#) · [PostgreSQL](#) · [Redis](#) · [YOLOv10](#) · [EasyOCR](#) · [RapidFuzz](#) · [Groq Llama 3.1](#) · [Git](#)

- Ranked Top 10, HackArena 2.0 Zonal Round. Built real-time barcode fraud detection (YOLOv10, EasyOCR, Redis, Groq Llama).
- Built fraud engine processing scans in <1s: inventory cross-reference, scan-frequency analysis, decision-tree risk scorer.
- Integrated YOLOv10 + EasyOCR + RapidFuzz for label-swap detection; auto-blocked transactions <65% trust with Llama fraud narratives.
- Engineered Redis UID uniqueness gate + NX locks to prevent double-scanning; WebSocket UI updates, SendGrid escalation.

MediHub — AI Prescription Manager

github.com/Debarghyasg/Medi-Hub[Node.js](#) · [Express](#) · [PostgreSQL](#) · [Flask](#) · [REST APIs](#) · [MobileNetV2 \(CNN\)](#) · [EasyOCR](#) · [KNN \(Scikit-Learn\)](#) · [Nodemailer](#) · [Node-Cron](#) · [Git](#)

- Developed end-to-end AI prescription digitization pipeline: CNN (MobileNetV2), EasyOCR, KNN classification, Node-Cron reminders.
- Built automated reminder system (Node-Cron + Nodemailer): daily DB scan, 24-hour pre-visit email alerts.
- Delivered secure multi-user platform (Bcrypt + Express-Session), REST API, real-time health dashboard; sole developer end-to-end.

PROJECT EXPERIENCE

Mahakash Club Website — Tech Lead

[Space Club](#)[React.js](#) · [Node.js](#) · [Express.js](#) · [PostgreSQL](#) · [HTML](#) · [CSS](#)

- Built full-stack event management platform for 500+ users: event registration, certificate generation, and archive of past ventures, videos, and blogs.

MSA GNIT Website — Full-Stack Developer

[Student Chapter](#)[React.js](#) · [Node.js](#) · [Spring Boot](#) · [PostgreSQL](#) · [HTML](#) · [CSS](#)

- Built full-stack tech-news and event platform for 1,000+ members: student blog, event registration, and certificate generation.

TECHNICAL SKILLS

Languages: Java (DSA), Python, C++, C, JavaScript, TypeScript, SQL | **Frameworks:** Next.js, FastAPI, Express.js, Flask, React**AI / ML:** LLM APIs (Groq, OpenAI), RAG pipelines, Sentence Transformers, YOLOv8/v10, EasyOCR, FinBERT, Presidio, Scikit-Learn, TensorFlow**Infrastructure:** Git, GitHub Actions (CI/CD), Docker, PostgreSQL, Redis, Celery, AWS (S3/SES/DynamoDB), Clerk Auth, SendGrid, Node-Cron, Alembic, pytest (Unit Testing)**Concepts:** Data Structures & Algorithms, OOP, System Design, REST APIs, Microservices, Distributed Systems, Scalability, Agile/Scrum, DBMS, Operating System, Computer Networking

LEADERSHIP & EXTRACURRICULARS

Governor — Mahakash Club (Space Club, GNIT)

Present

- Leading GNIT's space & astronomy club — organizing workshops, stargazing, and outreach programs.

Cloud Computing Head — Microsoft Student Associates (MSA), GNIT

2025-2026

- Leading Cloud Computing for MSA chapter — Azure sessions, cert prep, and hands-on labs.
- Collaborated with another leads to make it more engaging and user-friendly.

CERTIFICATIONS

- Oracle Cloud Infrastructure 2025 Certified Developer Professional — Oracle | OCI 2025 Certified Generative AI Professional — Oracle
- Oracle AI Autonomous Database 2025 Certified Professional — Oracle | Azure Management Tasks — Microsoft Learn | Fundamentals of Cybersecurity — Zscaler

EDUCATION

B.Tech / B.E. in Computer Science

Guru Nanak Institute of Technology, Kolkata

2023 – 2027 (Expected) | CGPA: 8.36 (through 5th Semester)